

growth in demand and importance of electronics as well as revolutionary changes in technology, launch of innovative products and the challenge of global competition. This has necessitated the electronic product and component manufacturers to focus on continuous improvement in their products in order to stay ahead of the pack, resulting in the following trends:

#### Convergence of technologies.

Convergence allows a single device to use multiple technologies/services. Convergence of technologies has become a reality in the last couple of years with the launch of main-line products enabling it. A smartphone is one of the key examples of the advent of convergence as it allows communication and computing to use the same device.

Convergence is moving beyond mobile phones to several other electronic devices like music players, IPTV, iPads and the like. This trend is expected to convert most electronic products into multi-utility products, thereby requiring high-tech electronic components and technology.

**Miniaturisation.** This refers to the creation of smaller-scale devices or components for mechanical, optical and electronic products. The dawn of convergence has led manufacturers to integrate multiple devices. At the same time, demand from consumers to reduce the size of products to make these easy to manage has led to these products becoming smaller in size or miniaturised.

Miniaturisation results in greater density of components that is usually possible through very-large-scale integration designs. This also enables lower cost of production, resulting in reduction in overall product price.

### Components manufacturing hubs

Most of the large electronic components manufacturing companies (Indian or multinational) are located around six cities of the country, namely, Delhi-NCR, Mumbai, Bengaluru, Hyderabad, Chennai and Pune. Delhi-NCR (which includes Noida, Gurgaon, Ghaziabad and Faridabad region) is the largest manufacturing hub, as it hosts a large number of component manufacturers. This is followed by Mumbai and Bengaluru. Apart from these, cities like Ahmedabad, Pune and Baroda are some of the upcoming manufacturing hubs in India (Fig. 4).

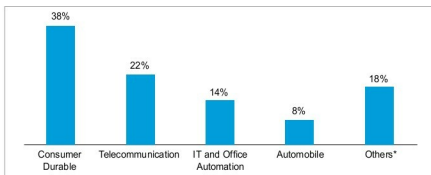


Fig. 3: Electronic components market share (Source: ELCINA study)

ing. Miniaturisation is expected to increase and impact the traditional component market as most of the traditional components will be replaced by chip components and integrated circuits.

**Artificial intelligence.** Consumers are becoming increasingly technology-conscious and are demanding products with built-in intelligence. This is resulting in electronics and consumer durable products being manufactured with intelligent functions and logic. For example, washing machines available now can sense the load and decide the appropriate washing cycles. Intelligence has moved beyond consumer products, and is also available in several medical and industrial electronics products with computer-numerical-control functions.

#### Moving forward

Above changes in technology are expected to change the landscape for electronic components manufacturing. Apart from the above, manufacturers across the globe have started moving towards green electronics and sustainable development with the implementation of Restriction of Hazardous Substances (RoHS)

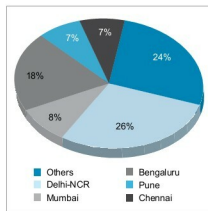


Fig. 4: Share of component manufacturing hubs in India (Source: ELCINA study)

and Waste Electrical and Electronic Equipment (WEEE) regulations.

In line with this trend, the Indian government, too, has issued notifications to regulate the use of hazardous substances (like lead, cadmium and mercury) and proper disposal of WEEE. Similarly, rules have also been notified with respect to energy-efficiency norms and labelling of most consumer electronics and durables.

Traditional components will increasingly face stiff competition from passive chip components, integrated circuits and surface-mount technologies that will replace these in several electronic products. Therefore in the near future, it is important for component manufacturers to shift focus from discrete components manufacturing to integrated components manufacturing so as to be able to benefit from the changing landscape of the country. **EFY**